

Data Collection and Preprocessing Phase

Date	4 October 2026
Team ID	N/A
Project Title	Student Mental Health Analysis (Tableau Dashboards & Story)
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	The dataset has 1,000 student records and 16 fields: Student_ID , Age (18 to 25), Gender (Male 341, Female 331, Other 328), Stress , Anxiety and Depression levels (1 to 10), Heart_Rate_Variability , Sleep_Duration_Hours , Attendance_Rate (%), Study_Hours_Per_Day , LMS_Activity_Score , Social_Interaction_Score , Academic_Performance_Index (50 to 100), Emotional_Journal (10 distinct sentences), Mental_Health_Risk (Low 414, Medium 347, High 239) and Personalized_Intervention_Strategy (8 strategies).
Data Cleaning	All 16 columns were checked: 0 missing values, 0 duplicate rows, Student_ID is unique, and the text columns have no stray spaces or spelling variants. So no rows were dropped or filled in. 14 outlier values (12 sleep, 2 heart rate variability) were kept because they are plausible.
Data Transformation	Done in Tableau: Academic_Performance_Index grouped into grade bands (A+, A, B, C, F) for the grade pie chart; Study_Hours_Per_Day grouped into 1-hour buckets; counts turned into percentages of students for the risk mix by gender; average KPI tiles (mental score, sleep, depression, attendance) with a Gender filter; risk levels coloured High, Medium and Low.
Data Type Conversion	All numeric columns were already numbers and the text columns already text, so no errors had to be fixed. Age was set as a Dimension (category) so it can be used in Age vs Risk, and Student_ID was kept as an identifier rather than a measure to total.

Column Splitting and Merging	No splitting or merging was needed because each column holds one value. Emotional_Journal has only 10 distinct sentences, so it was used as a category as it is.
Data Modeling	The data is a single table, so no joins or relationships are needed. Measures were built in Tableau: averages of stress, depression, heart rate variability, sleep and attendance by gender, record count and percentage of students, and trend lines (study hours vs academic score $R^2 \approx 0$, $p = 0.84$; sleep vs stress $R^2 = 0.0006$).
Save Processed Data	The raw file needed no cleaning edits, so it is kept unchanged as the Tableau data source. All calculated fields, dashboards and the Story are saved in the Tableau workbook, which is published to Tableau Public for the Flask web integration.